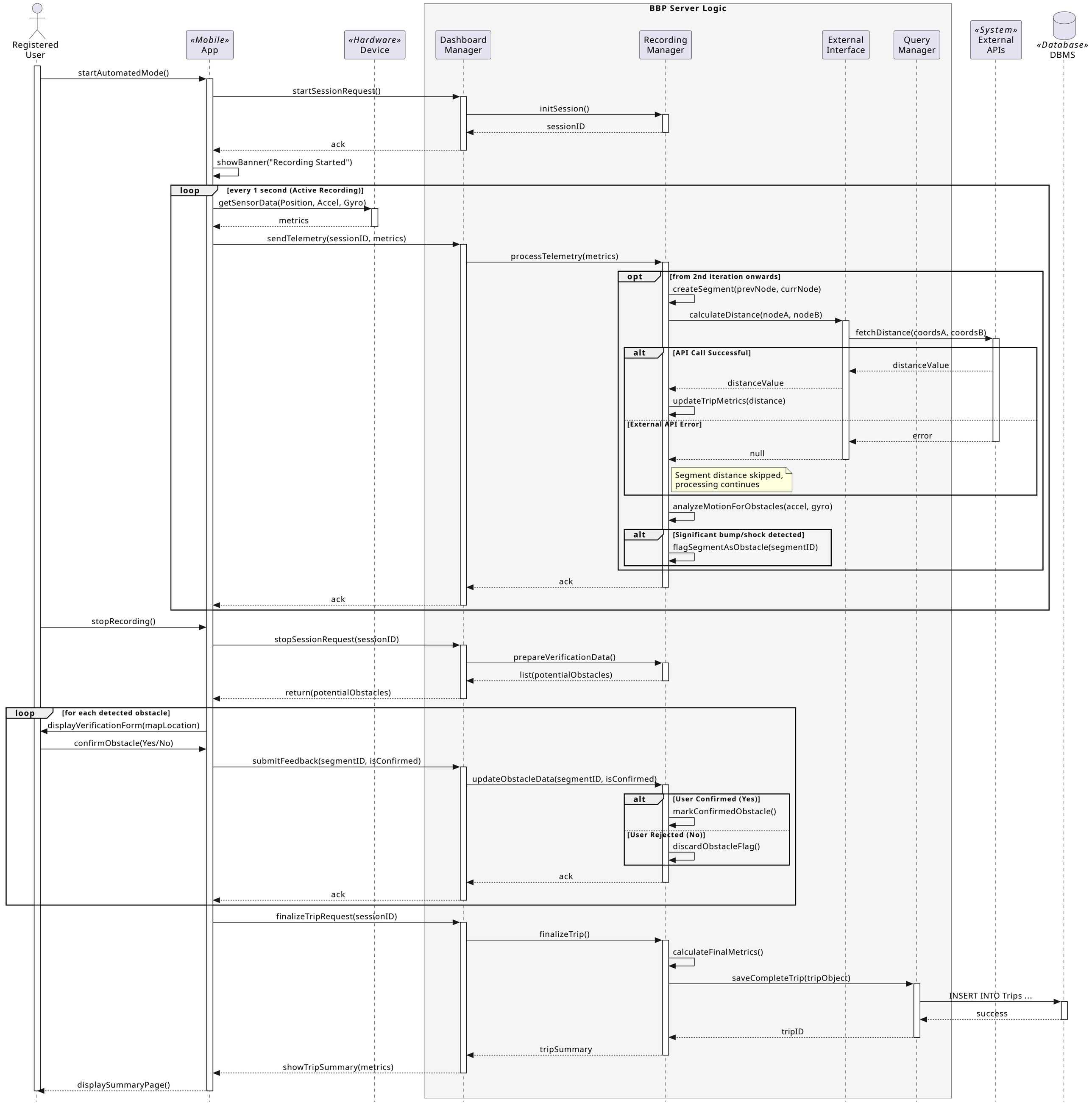
BBP Server Logic
Registered User
«Mobile» App
«Hardware» Device
Dashboard Manager
Recording Manager
External Interface
Query Manager
«System» External APIs
«Database» DBMS
startAutomatedMode()
startSessionRequest()
initSession()
sessionID
ack
showBanner("Recording Started")
loop
[every 1 second (Active Recording)]
getSensorData(Position, Accel, Gyro)
metrics
sendTelemetry(sessionID, metrics)
processTelemetry(metrics)
opt
[from 2nd iteration onwards]
createSegment(prevNode, currNode)
calculateDistance(nodeA, nodeB)
fetchDistance(coordsA, coordsB)
alt
[API Call Successful]
distanceValue
distanceValue
updateTripMetrics(distance)
[External API Error]
error
null
Segment distance skipped, processing continues
analyzeMotionForObstacles(accel, gyro)
alt
[Significant bump/shock detected]
flagSegmentAsObstacle(segmentID)
ack
ack
ack
stopRecording()
stopSessionRequest(sessionID)
prepareVerificationData()
list(potentialObstacles)
return(potentialObstacles)
loop
[for each detected obstacle]
displayVerificationForm(mapLocation)
confirmObstacle(Yes/No)
submitFeedback(segmentID, isConfirmed)
updateObstacleData(segmentID, isConfirmed)
alt
[User Confirmed (Yes)]
markConfirmedObstacle()
[User Rejected (No)]
discardObstacleFlag()
ack
ack
finalizeTripRequest(sessionID)
finalizeTrip()
calculateFinalMetrics()
saveCompleteTrip(tripObject)
INSERT INTO Trips ...
success
tripID
tripSummary
showTripSummary(metrics)
displaySummaryPage()